A MINI PROJECT REPORT

ON

FOOD WASTE DASHBOARD USING PYTHON

Restaurant Demand, Waste & Cost Analytics

Submitted in partial fulfillment of the requirements for

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING (AI / AIML)

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ACADEMIC YEAR 2026-2027

CERTIFICATE

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This is to certify that the mini project report titled **“FOOD WASTE DASHBOARD USING PYTHON”** is prepared by the following students:

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The report is submitted to the Department of Computer Science and Engineering (AI / AIML), Dr. K. V. Subba Reddy Institute of Technology, Kurnool, in partial fulfillment of the requirements for the Bachelor of Technology mini-project academic activity.

Guide: **Dr. B. Mahesh**, Associate Professor, Department of CSE. The accompanying chapters describe the restaurant food waste application, dataset, visualizations and demand estimation method.

Guide and departmental signatures are to be added after academic review and approval.

Project Guide	Head of Department
Signature: _____	Signature: _____

DECLARATION

We declare that this mini project report entitled **“FOOD WASTE DASHBOARD USING PYTHON”** describes the design and functionality of the supplied Restaurant Food Waste Streamlit project.

The technical descriptions are based on app.py, restaurant_food_waste_dataset.csv, requirements.txt and README.txt in the uploaded project archive. Calculated results in this report are derived from the supplied CSV.

The dataset is synthetic and intended for educational demonstration. Reported totals represent simulated records, not verified restaurant operations. The demand estimator is a historical-median baseline and is not a trained machine-learning model.

We understand the limitations described in the report and will review the final submission before signing.

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ACKNOWLEDGEMENT

We express our sincere gratitude to the management and faculty of Dr. K. V. Subba Reddy Institute of Technology, Kurnool, for providing an academic environment that encourages practical learning and project development.

We acknowledge Dr. B. Mahesh, Associate Professor, Department of CSE, whose guidance details are retained from the supplied academic report format.

We thank our department teachers, classmates and family members for their encouragement during the preparation and documentation of this Food Waste Dashboard project.

This work demonstrates the use of Python, Streamlit, Pandas, NumPy and Plotly to turn meal-service records into interactive summaries, waste indicators and a transparent demand-estimation demonstration.

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REPORT GUIDE / TABLE OF CONTENTS

Section	Title	Page
01	Abstract	6
02	Introduction	6
03	Literature Survey / Existing System	7
04	Technologies and Tools Used	8
05	System Requirements	9
06	Dataset Description	10-11
07	Implementation	12-13
08	Data Visualization	14-15
09	Results and Discussion	16
10	Advantages and Limitations	17
11	Conclusion	17
12	Future Enhancement	18
13	References	18
14	Appendix	19-24

Evidence guide

Source review confirms application features. Dataset calculations reproduce report totals. Printed charts are analytical figures generated from the supplied data. Appendix C contains four user-provided application screenshots. Proposed improvements are identified separately.

01 Abstract

The Food Waste Dashboard is an interactive restaurant analytics application built with Python and Streamlit. It helps users explore meals prepared, meals sold, unsold meals and estimated waste costs. Sidebar filters select meal type, special event and restaurant type. Four tabs organize overview charts, demand patterns, waste and cost analysis, and a simple demand estimation demonstration.

The project contains 1,200 synthetic records covering dates from 2 January to 31 December 2025. Across all records, 64,045 meals were prepared, 54,692 were sold and 9,353 were wasted. The aggregate waste rate is 14.60%, with an estimated waste cost of INR 860,929.79. These are dataset totals; the current default filter displays a smaller subset, explained in Sections 6 and 9.

The baseline estimator uses the median sales of matching historical records and adds a user-selected preparation buffer. The application supports CSV upload, interactive Plotly charts and filtered CSV download.

Keywords: food waste; restaurant analytics; Streamlit; Pandas; Plotly; demand estimation; synthetic data.

02 Introduction

2.1 Problem statement

Develop an interactive dashboard that summarizes restaurant meal preparation and sales, reveals patterns in unsold food and estimated costs, and demonstrates a transparent method for planning preparation quantities.

2.2 Objectives and scope

• Monitor preparation, sales, waste and estimated costs through five KPI cards.

• Compare meal services, weekdays, menu categories and event groups.

• Explore uploaded CSV records and export the selected subset.

• Demonstrate historical-median demand estimation with a preparation buffer.

03 Literature Survey / Existing System

3.1 Review of common approaches

This section compares common workflow types. It is a conceptual comparison, not a formal empirical literature review.

Approach	Typical role	Limitation
Manual daily registers	Record preparation and sales	Repeated manual calculation and limited visual comparison
Spreadsheet reports	Summarize totals with formulas and charts	Formulas and filter rules require careful maintenance
Point-of-sale records	Capture transactions and demand history	Preparation and discarded-food data may be separate
This project	Combine CSV summaries, charts and a baseline estimator	Synthetic data; no database or operational integration

3.2 Gap identified

A classroom project needs a clear workflow from restaurant records to interpretable indicators. A tabbed dashboard makes it easier to inspect the relationship between preparation and sales, compare waste categories and discuss how preparation buffers affect planning.

3.3 Proposed system

The supplied system loads a CSV, parses the date field, builds sidebar filter options and summarizes matching rows. Plotly figures display comparisons and distributions. A separate estimator uses the complete loaded dataset to locate comparable records and calculate a median demand value.

3.4 Intended users

Students and project evaluators can use it to learn data visualization. Restaurant planners could explore the approach after replacing the sample data, resolving data-quality issues and validating results against actual operations.

04 Technologies and Tools Used

Technology	Version requirement	Role
Python	3.10+ recommended in README	Application execution
Streamlit	>=1.35	Web interface, filters, tabs, uploads and downloads
Pandas	>=2.0	CSV reading, dates, filtering and aggregation
NumPy	>=1.24	Ceiling operation for buffered preparation
Plotly	>=5.20	Interactive line, bar, box, scatter and donut charts
CSV	17-column input file	Portable dataset storage

4.1 Project files

File	Purpose
app.py	Main dashboard and demand estimation logic
restaurant_food_waste_dataset.csv	1,200 synthetic restaurant records
requirements.txt	Four dependency requirements
README.txt	Project description, launch steps and limitations

4.2 Execution model

Streamlit runs a local Python service and displays its interface in a browser. Widgets trigger script reruns. The load_data function uses st.cache_data to cache CSV loading. Pandas computes summaries and Plotly figures are embedded with st.plotly_chart.

4.3 Data storage

The supplied project reads a local CSV or an uploaded CSV. It does not contain a database, login system, persistent transaction entry form or trained model file.

05 System Requirements

Component	Suggested setup
Computer	Modern Windows, macOS or Linux laptop
Memory	4 GB or more recommended for comfortable browser use
Python	Python 3.10 or newer, as specified in the project README
Browser	A current browser capable of displaying Streamlit
Internet	Needed to obtain Python packages; no external data API is used
Project folder	Keep app.py, the CSV and requirements.txt together

5.1 Run on a Windows laptop

• Extract the supplied ZIP into a normal folder.

• Open Command Prompt in the extracted folder.

• Install the required packages and launch the dashboard with the commands below.

Command	Purpose
python --version	Check the Python installation
python -m pip install -r requirements.txt	Install dependencies
python -m streamlit run app.py	Start the dashboard

Open the local URL printed by Streamlit, normally http://localhost:8501. Keep the terminal running while using the dashboard. Press Ctrl+C in the terminal to stop it.

5.2 Troubleshooting

If the dataset is not found, place it beside app.py and start from that folder, or upload it in the sidebar. If the default port is occupied, append --server.port 8502 to the launch command. On systems using the python3 command, substitute python3 for python.

06 Dataset Description

6.1 Source and coverage

The bundled CSV is a synthetic educational dataset with 1,200 rows and 17 columns. Its dates range from 2025-01-02 to 2025-12-31. A row represents a simulated meal-service record, not a uniquely identified restaurant or a complete daily census.

Column	Meaning
date; day_of_week	Service date and weekday label
month; is_weekend	Month number and weekend indicator
meal_type	Breakfast, Lunch or Dinner
restaurant_type	Cafe, Casual Dining, Family Restaurant, Quick Service
menu_category	Chinese, Desserts, North Indian, Snacks, South Indian
weather_temperature_c	Simulated temperature in degrees Celsius
rainfall_mm	Simulated rainfall in millimetres
special_event	Festival, Local Event, Promotion or literal None
restaurant_capacity	Capacity associated with the record
meals_prepared; meals_sold	Prepared quantity and sold quantity
food_waste_meals	Count of unsold meals
ingredient_waste_kg	Simulated ingredient waste in kilograms
estimated_cost_per_meal_inr	Estimated cost per meal in Indian rupees
estimated_waste_cost_inr	Estimated cost of wasted meals in Indian rupees

The file does not include a restaurant ID, ingredient-level inventory, customer identity or an independently observed demand target.

06 Dataset Description / Data Quality

6.2 Checks performed on the supplied file

Check	Result
Row and column count	1,200 rows; 17 columns
Duplicate complete rows	0
Date parsing	All dates parse successfully
Meal balance	Prepared minus sold equals wasted meals for all 1,200 rows
Waste cost relationship	Rounded wasted meals × unit cost matches stored waste cost
Event parsing with default Pandas settings	897 literal None entries are read as missing values

6.3 Effect of the event parsing issue

The source calls pd.read_csv with its default missing-value handling. The literal text None becomes a missing value. Event options are then created using dropna(), and the default filter selects only those options. Consequently, 897 no-event records are excluded and only 303 records appear in the default analytics view.

6.4 Recommended correction

A future code correction can preserve the event label when reading CSV, for example with keep_default_na=False, followed by explicit numeric and date validation. Another option is to fill missing special_event values with a clear No Event label before building filters. This report documents the uploaded code without modifying it.

6.5 Interpretation of units

Meal waste is a count; ingredient waste is measured in kilograms. These quantities should not be added. Estimated waste cost is derived from meal counts and unit costs, and does not represent measured revenue loss or total business profit.

07 Implementation

7.1 Application workflow

Stage	Processing
1. Input	Load bundled CSV or sidebar upload
2. Preparation	Parse date using errors="coerce"
3. Selection	Filter meal type, event and restaurant type
4. Aggregation	Compute KPI totals, means and grouped summaries
5. Presentation	Render four Streamlit tabs and Plotly figures
6. Export	Download all filtered rows as CSV
7. Estimation	Use complete loaded data for matching and median demand

7.2 KPI calculations

Indicator	Formula
Meals prepared	Sum of meals_prepared in filtered rows
Meals sold	Sum of meals_sold in filtered rows
Meals wasted	Sum of food_waste_meals in filtered rows
Waste rate (%)	100 × total wasted / total prepared; zero if denominator is zero
Estimated waste cost	Sum of estimated_waste_cost_inr in filtered rows

7.3 Interface and export behavior

The Overview tab shows a preview of the first 100 filtered rows. The download button exports the complete filtered selection. If no rows match, the app displays a warning and stops further rendering. If the local dataset is absent, a specific error message explains how to supply it.

07 Implementation / Demand Estimation

7.4 Matching procedure

• Select meal type, weekday, event and restaurant type.

• Find rows matching all four categorical selections in the complete loaded dataset.

• If fewer than five rows match, broaden to meal type plus restaurant type.

• If no rows remain, broaden again to the selected meal type.

• Round the median meals_sold to estimate demand.

• Apply a 0-25% buffer and round upward to obtain suggested preparation.

7.5 Formulas

Output	Calculation
Estimated demand	int(round(median(meals_sold)))
Suggested preparation	ceil(estimated demand × (1 + buffer / 100))
Matched historical waste rate	100 × matched wasted meals / matched prepared meals

7.6 Important implementation details

The temperature slider is displayed but its value is not used in matching or calculation. The estimator uses df rather than filtered, so sidebar selections do not restrict its input records. The method does not compute distances, train a model or use rainfall as a predictor.

7.7 Reproducible example

For Dinner, Sunday, Festival and Casual Dining, the strict match has two rows. The code therefore falls back to all Dinner / Casual Dining rows (116 records). The median-based estimate is 44 meals; a 5% buffer gives 47 suggested meals. This illustrates the fallback logic, not forecasting accuracy.

08 Data Visualization

All figures in this section use the 303 rows included by the current default filters. They are print equivalents derived from the CSV, not browser screenshots. Plotly charts in the application support interactive inspection.

8.1 Grouped bar comparison

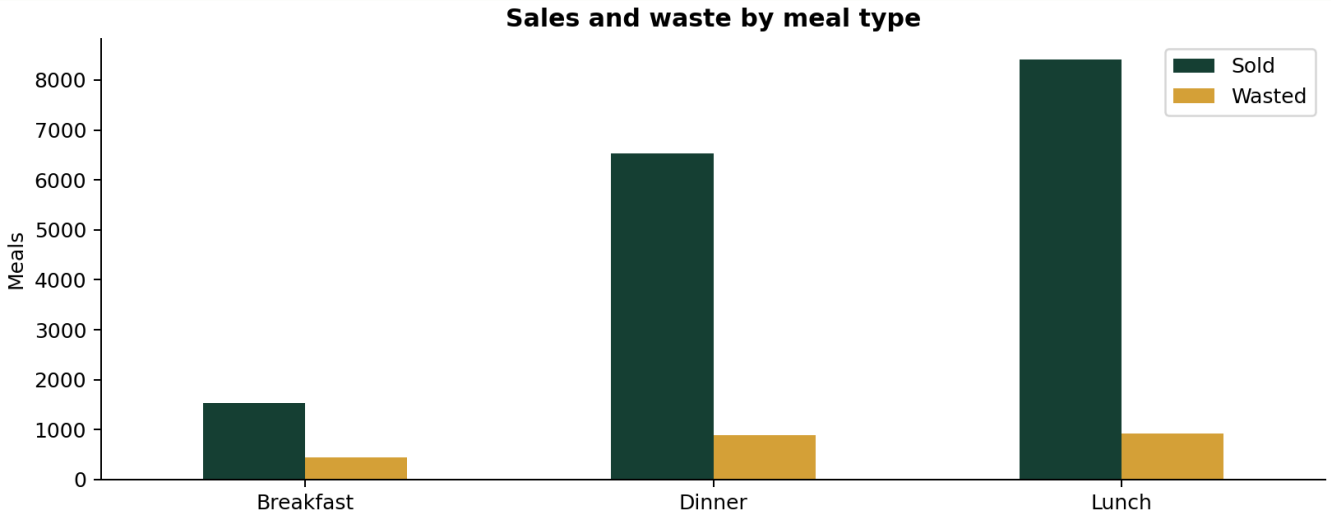


Figure 1. Sold and wasted meals by meal type in the default selected subset.

8.2 Donut chart

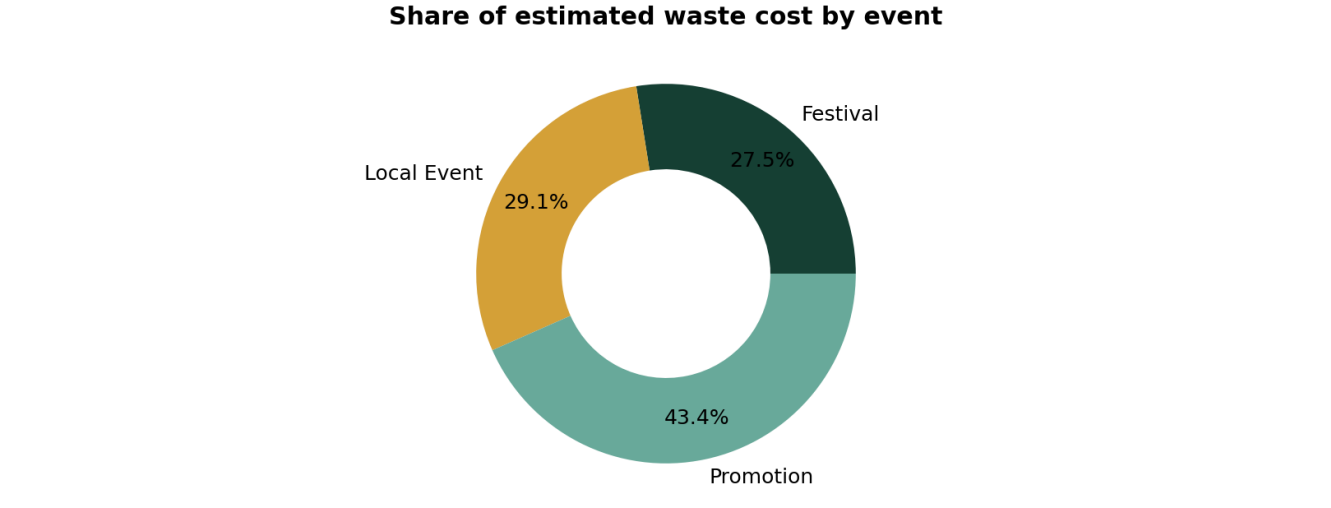


Figure 2. Event-wise share of estimated waste cost; no-event rows are excluded by the current parser/filter behavior.

08 Data Visualization / Trends and Categories

8.3 Time-series interpretation

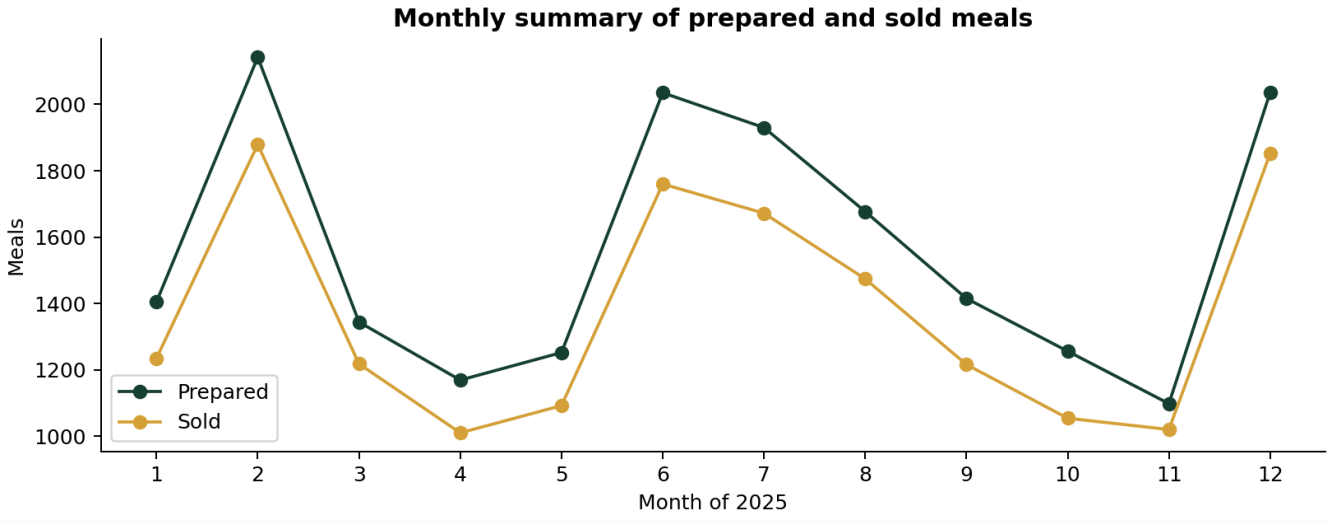


Figure 3. Monthly print summary of the selected records. The application itself groups its line chart by individual date.

The gap between prepared and sold quantities summarizes unsold meals. Uneven numbers of records across periods can influence the totals, so these synthetic patterns should not be interpreted as market forecasts.

8.4 Menu category comparison

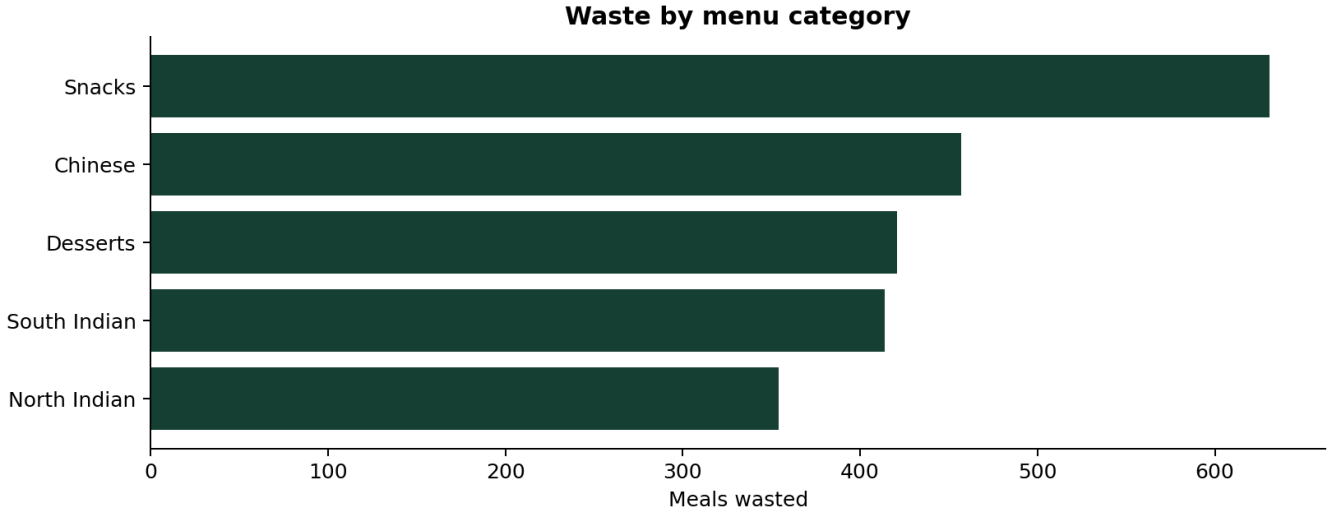


Figure 4. Aggregate wasted meals by menu category in the selected subset.

The application also includes weekday average-demand bars, meal-type box plots and a temperature-versus-sales scatter plot sized by restaurant capacity. These describe the supplied records and do not establish causal relationships.

09 Results and Discussion

9.1 Recomputed results

The following values were calculated directly from the bundled CSV using the application's aggregation rules. Default-view results include only rows with non-missing parsed events.

Metric	All supplied records	Default selected subset
Records	1,200	303
Meals prepared	64,045	18,759
Meals sold	54,692	16,482
Meals wasted	9,353	2,277
Waste rate	14.60%	12.14%
Estimated waste cost (INR)	860,929.79	215,011.35

9.2 Meal-type results in the default subset

Meal type	Sold	Wasted
Breakfast	1,538	451
Dinner	6,533	895
Lunch	8,411	931

9.3 Discussion

Lunch contributes the largest sold-meal and wasted-meal totals in the default subset. Higher absolute waste does not by itself imply poorer efficiency: meal volume and the number of records should also be considered. Compare waste rates alongside counts before drawing operational conclusions.

9.4 Verification scope

Source code and dataset relationships were inspected, and arithmetic was recomputed. The Streamlit interface was not executed for this report. No forecasting accuracy, restaurant savings, user-study result or production performance is claimed.

10 Advantages and Limitations

10.1 Advantages

• A compact project with clear dependencies and a familiar browser interface.

• Five headline indicators and multiple interactive chart types.

• CSV upload and export for classroom exploration of different datasets.

• Transparent median estimation and a user-controlled preparation buffer.

10.2 Limitations

• Synthetic data cannot establish actual restaurant savings or prediction quality.

• Default event parsing excludes 897 no-event rows from analytics.

• Temperature control does not affect the estimator.

• No model training, backtesting, uncertainty interval or forecast horizon.

• Uploaded data lack full schema and numeric validation; malformed files can fail.

• No authentication, database, inventory integration or durable data-entry workflow.

11 Conclusion

The Food Waste Dashboard demonstrates how a Python data-visualization project can combine meal-service records, summary indicators, comparisons and a simple preparation-planning example. Streamlit provides the interface, Pandas prepares the data and Plotly supports interactive exploration.

The project is suitable for academic demonstration when the synthetic-data scope and event-filter issue are clearly explained. Improving input validation and assessing the baseline against genuine historical records are the next steps toward operational usefulness.

12 Future Enhancement

Proposed enhancement	Expected benefit
Repair event handling	Include no-event records under an explicit label
Validate uploaded CSVs	Check required columns, types, ranges and missing values
Add restaurant IDs and date filters	Enable consistent restaurant and period comparisons
Use evaluated forecasting models	Compare against the median baseline using held-out time periods
Add uncertainty and service targets	Show trade-offs between shortages and over-preparation
Connect sales and inventory systems	Reduce manual entry and improve record completeness
Add secure storage and user roles	Support controlled access and lasting records

13 References

Primary materials examined for this report:

• restaurant_food_waste_streamlit_project.zip: supplied project archive.

• app.py: Streamlit layout, filtering, chart definitions and demand-estimation logic.

• restaurant_food_waste_dataset.csv: synthetic sample records used for all calculations and printed figures.

• requirements.txt and README.txt: declared package requirements and launch instructions.

• Smart_Crop_Dashboard.pdf: supplied academic structure, institutional details, logo and visual style reference.

No external restaurant dataset, published accuracy claim or field study is used as evidence in this report.

14 Appendix

Appendix A. Source-code structure

Block	Responsibility
Imports and page configuration	Load libraries; configure wide layout and page title
load_data	Read CSV and parse dates with Streamlit caching
Sidebar and filtered frame	Build category choices and filter rows
KPI calculations	Compute five headline metrics
Overview tab	Daily trend, meal bars, preview and CSV export
Demand patterns tab	Weekday bars, box plot and temperature scatter
Waste & cost tab	Category bars, event donut and average metrics
Prediction demo tab	Categorical matching, median estimate and buffer

Appendix B. Suggested acceptance checks

Check	Expected outcome / status
Default bundled data	303 rows selected with the current parsing behavior
Clear any sidebar selection	No matching rows warning; no charts rendered
Filtered CSV download	All selected rows exported; preview limited to 100
5% buffer example	Preparation equals ceiling of demand × 1.05
Change temperature only	Current source produces no estimator change
Malformed CSV upload	Needs explicit validation in a future revision

These are code-derived acceptance expectations; they are not a record of completed browser-based tests.

14 Appendix

3. C. Additional Screenshots

C.1 Overview dashboard

The overview screen shows preparation, sales and waste KPIs, dashboard filters, and the prepared-versus-sold and meal comparison charts.

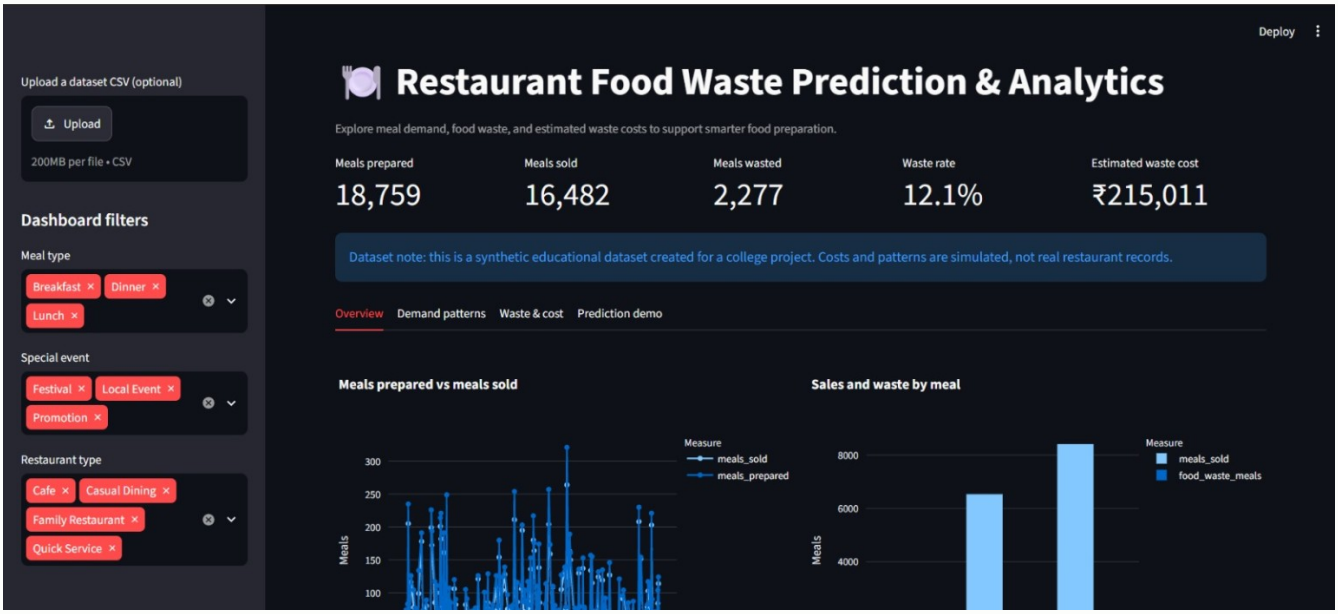


Figure C.1. User-provided screenshot of the Restaurant Food Waste Dashboard.

The screenshot is reproduced at its original aspect ratio. Any content cut off at the screen edges was already outside the supplied image.

14 Appendix

3. C. Additional Screenshots

C.2 Demand patterns

The demand patterns screen shows average meal demand by weekday and the distribution of meals sold by meal type.

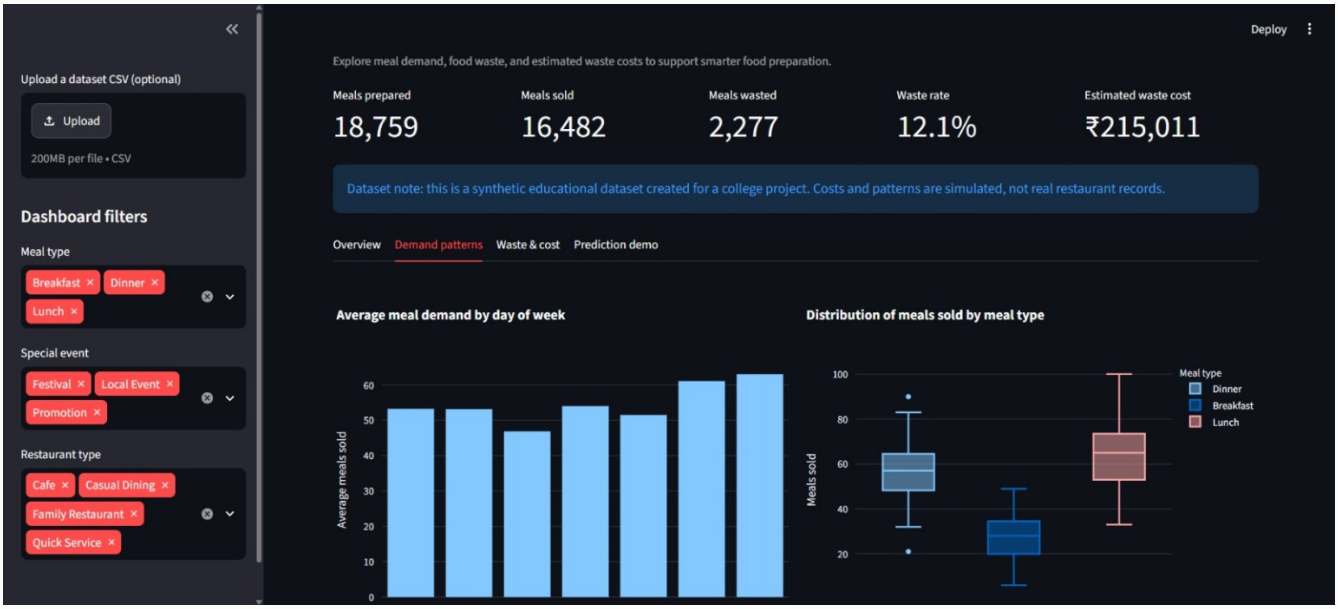


Figure C.2. User-provided screenshot of the Restaurant Food Waste Dashboard.

The screenshot is reproduced at its original aspect ratio. Any content cut off at the screen edges was already outside the supplied image.

14 Appendix

3. C. Additional Screenshots

C.3 Waste and cost analysis

The waste and cost screen compares wasted meals by menu category and the share of estimated waste cost by event.



Figure C.3. User-provided screenshot of the Restaurant Food Waste Dashboard.

The screenshot is reproduced at its original aspect ratio. Any content cut off at the screen edges was already outside the supplied image.

14 Appendix

3. C. Additional Screenshots

C.4 Prediction demonstration

The prediction screen provides meal, weekday, event and restaurant controls, together with temperature and preparation-buffer sliders.

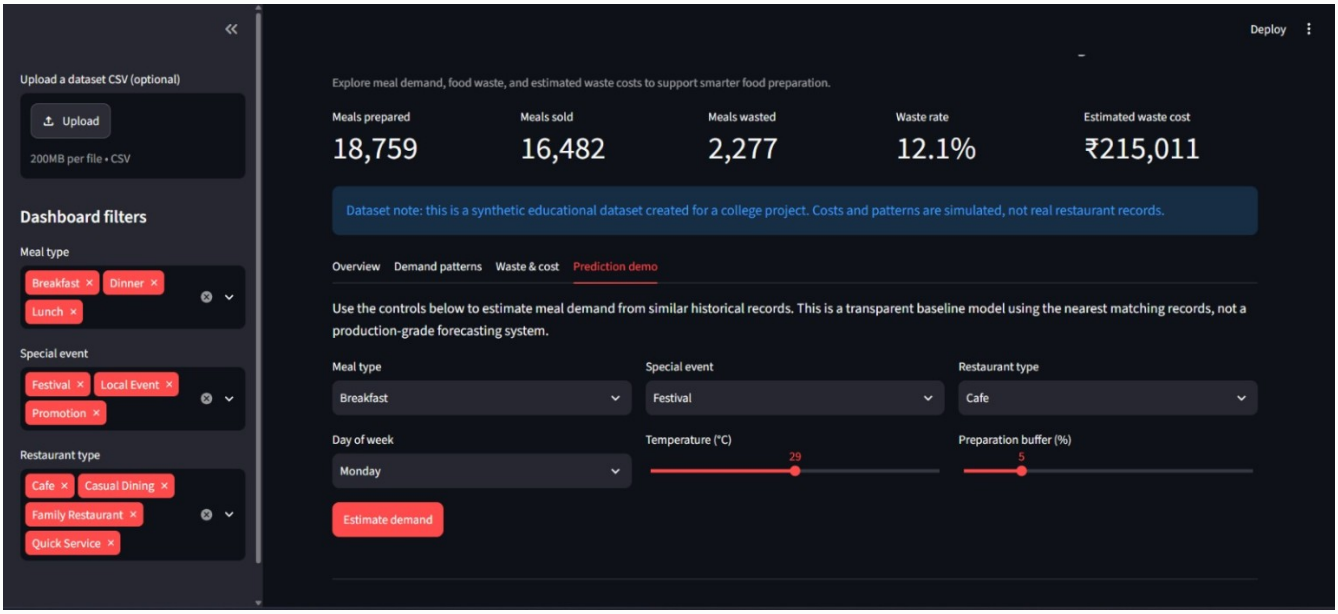


Figure C.4. User-provided screenshot of the Restaurant Food Waste Dashboard.

The screenshot is reproduced at its original aspect ratio. Any content cut off at the screen edges was already outside the supplied image.

14 Appendix / Reproducibility

Appendix D. Chart and evidence register

Figure	Input and aggregation
1. Meal comparison	303 selected rows; sum sold and wasted by meal_type
2. Event cost share	303 selected rows; sum estimated waste cost by special_event
3. Monthly comparison	303 selected rows; sum prepared and sold by calendar month
4. Category waste	303 selected rows; sum wasted by menu_category

The four images in Appendix C were supplied by the user and show the food-waste application interface. The figures in Section 8 are separate analytical charts generated from the dataset. No simulated interface is presented as an actual screenshot.

Appendix E. Reproducing the reported totals

Read restaurant_food_waste_dataset.csv using Pandas. For full-file totals, sum the prepared, sold, wasted and estimated-cost columns without filtering. For default-view totals, retain rows whose parsed special_event is not missing. Calculate waste rate from the total wasted divided by the total prepared, multiplied by 100.

Appendix F. Submission checklist

• Keep the project ZIP and this report together for evaluation.

• Verify the five student names and roll numbers on the opening pages.

• Obtain guide and department approval before adding signatures.

• Demonstrate the four tabs, filters, CSV export and baseline estimator.

• Explain the sample-data scope and the documented event parsing limitation.

Prepared for academic year 2026-2027